

Abhimanue Udayan

AI Solutions Architect | Founder | Full-Stack AI & DevSecOps Engineer

@ abhimanueudayan@gmail.com +91 9496654440 LinkedIn abhimanue.in Kochi, India Age: 35

Founder and AI Solutions Architect with experience designing, building, and deploying production-ready AI SaaS platforms from concept to scale. Skilled in Django, React, FastAPI, PostgreSQL, Redis, Docker, AWS, CI/CD, and modern DevSecOps practices, with hands-on experience translating complex customer requirements into scalable cloud-based solutions. Experienced in delivering technical demonstrations, solution architecture, and enterprise AI implementations while collaborating directly with customers to drive product adoption.

In 2017, became one of the few people in the world to compute a fully verified 206,545-digit solution to the Archimedes' Cattle Problem (ACP), demonstrating expertise in algorithm design, mathematical optimization, and large-scale computational problem solving. Passionate about AI-assisted software development, developer productivity, and helping organizations accelerate software delivery through intelligent automation, modern DevSecOps, and cloud-native engineering.

Work Experience

Founder - CTO

Hexagon Know Private Limited

Jan 2024 – June 2026 Kochi, India

- Designed and built a production AI SaaS platform using Django, React, FastAPI, PostgreSQL, Redis, Celery, and AWS.
- Developed LMS solutions and AI video generation tools for businesses to train staff and produce demo videos.
- Integrated large language models to automate content generation, multimedia creation, and enterprise workflows.
- Worked directly with enterprise customers to prototype AI solutions, demonstrate technical capabilities, and customize deployments.
- Led end-to-end product architecture from concept to production deployment, including cloud infrastructure, APIs, authentication, and monitoring.

Associate - Curriculum Development- Physics

Aakash Byju's Foundation - BYJU'S - Think and Learn Pvt. Ltd.

Dec 2021- August 2023 Bangalore, India

- Led large-scale curriculum and content systems serving students across India, coordinating distributed teams and review workflows.
- Built structured content pipelines, simulations, and quality-control processes to improve learning delivery efficiency.
- Authored technical scripts and product-facing content for web and app experiences.

Academic Excellence Manager

LEAD School - Leadership Boulevard Pvt. LTD

Jan 2021- June 2021 Coimbatore, India

- Managed LMS content delivery across 20+ partner schools, acting as a bridge between product teams, educators, and end users.
- Led onboarding and implementation programs for 15 partner institutions across Kerala.
- Collaborated with sales and product teams, contributing to deal closures and long-term client engagement.

Key Skills

- Solution Architecture
- Full Stack Development
- AI-Assisted Development Workflows
- Enterprise Customer Engagement
- Rapid Prototyping & MVP Development
- Developer Productivity & Automation
- Technical Workshops & Enablement
- API Design & System Integration
- Cloud Architecture (AWS)
- Technical Consulting & Product Strategy

Technical Skills

Django	●●●●●
React Js	●●●●●
PostgreSQL	●●●●●
FastAPI	●●●●●
AWS & Cloud	●●●●●
Redis, Celery	●●●●●
REST APIs	●●●●●
System Security	●●●●●
OpenAI API	●●●●●
Docker	●●●●●
Prompt Engineering	●●●●●
LLM Integration	●●●●●

Soft Skills

- Training System Strategy & Product Thinking
- High Ownership & Execution
- Creative Problem Solving
- Clear Technical & Visual Communication
- Rapid Product Iteration
- Cross-Functional Collaboration
- Client-Focused Solution Development
- Feedback-Driven Improvement
- Systems Thinking & Operational Optimization

Senior Faculty- Science

Orchids The International School - K12 Techno Services

May 2018- Aug 2020 Bangalore, Karnataka India

- Taught physics & chemistry to students preparing for grade 8,9,10.
- Trained students for Olympiads, research, competitions, and academics.
- Developed textbooks and digital learning content for students.

Formal Education/Degree

M.Sc. Applied Physics

Visvesvaraya National Institute of Technology, Nagpur, India (VNIT)

August 2015 – September 2017 Nagpur, India

- Focused on computational systems, algorithm design, and large-scale numerical optimization.
- Built custom programs to solve complex mathematical problems requiring high-precision arithmetic and performance tuning.
- Solved the Archimedes' Cattle Problem with 100% accuracy calculating solutions of 18 million digit number. by developing an optimized computational approach for handling extremely large integer operations.

B.Sc. Physics

Sacred Heart College, Thevara

August 2009 – June 2012 Kochi, India

- Trained in mathematical physics, numerical programming, microprocessor programming, electronics, and design.
- Elected as chief editor of the college magazine and delivered the 2010-11 college magazine named "Version3.0 : India 2020" - inspired by the Scientist and 11th President of India, APJ Abdul Kalam's India 2020.

Computer Science - Higher Secondary Education

SNDP HSS Udayamperoor

2007 Kochi, India

SSLC

SNDP HSS Udayamperoor

August 2007 – May 2009 Kochi, India

Recent Events and Workshops

Moderator : Journey of Modern Entrepreneurship

TiE Global Summit

2026 Kochi, India

Moderated a panel of entrepreneurs and investors conducting an effective discussion on building the right team, finding the right idea, and getting the first customers.

Design Thinking and Business Plan Workshop

TiE Young Entrepreneurs

2025 Kochi, India

Led workshops across multiple schools in Ernakulam on entrepreneurship, design thinking, fundraising, and business plan development.

Achievements and Awards

IIT JAM : Joint admission test for M.Sc. admissions in IITs

IIT Guwahati

2015

India

- 94.8 Percentile

Elected as the Chief Editor of the College magazine

Sacred Heart College

2010

Kochi

- 1500+ votes

Chess Champion

Byju's, Sacred Heart College & More

2023, 2016, 2011

Bangalore, Kerala

- 1st place.

Recognition and Awards At The Workplace

AEM of the Month

LEAD School

May 2021

Coimbatore, India

Support Agent of the Month

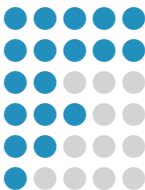
Sutherland Global Services

Oct 2013

Cochin, India

Languages

English
Malayalam
Sanskrit
Hindi
Tamil
Kannada



Hobbies

Chess
Driving
Teaching
Computer Gaming
Swimming
Trekking

